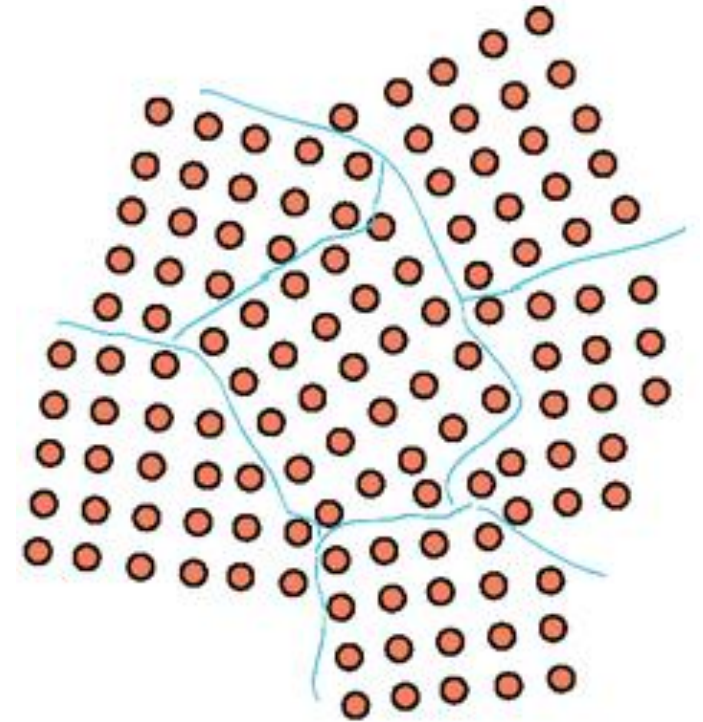


316-221 METALLURGY

CHAPTER 7: STRENGTHENING MECHANISMS OF METALS

CONTENTS

- ☐ Introduction
- ☐ Strain hardening
- ☐ Grain boundary strengthening
- ☐ Solid solution strengthening
- ☐ Precipitation hardening
- ☐ Dispersion strengthening



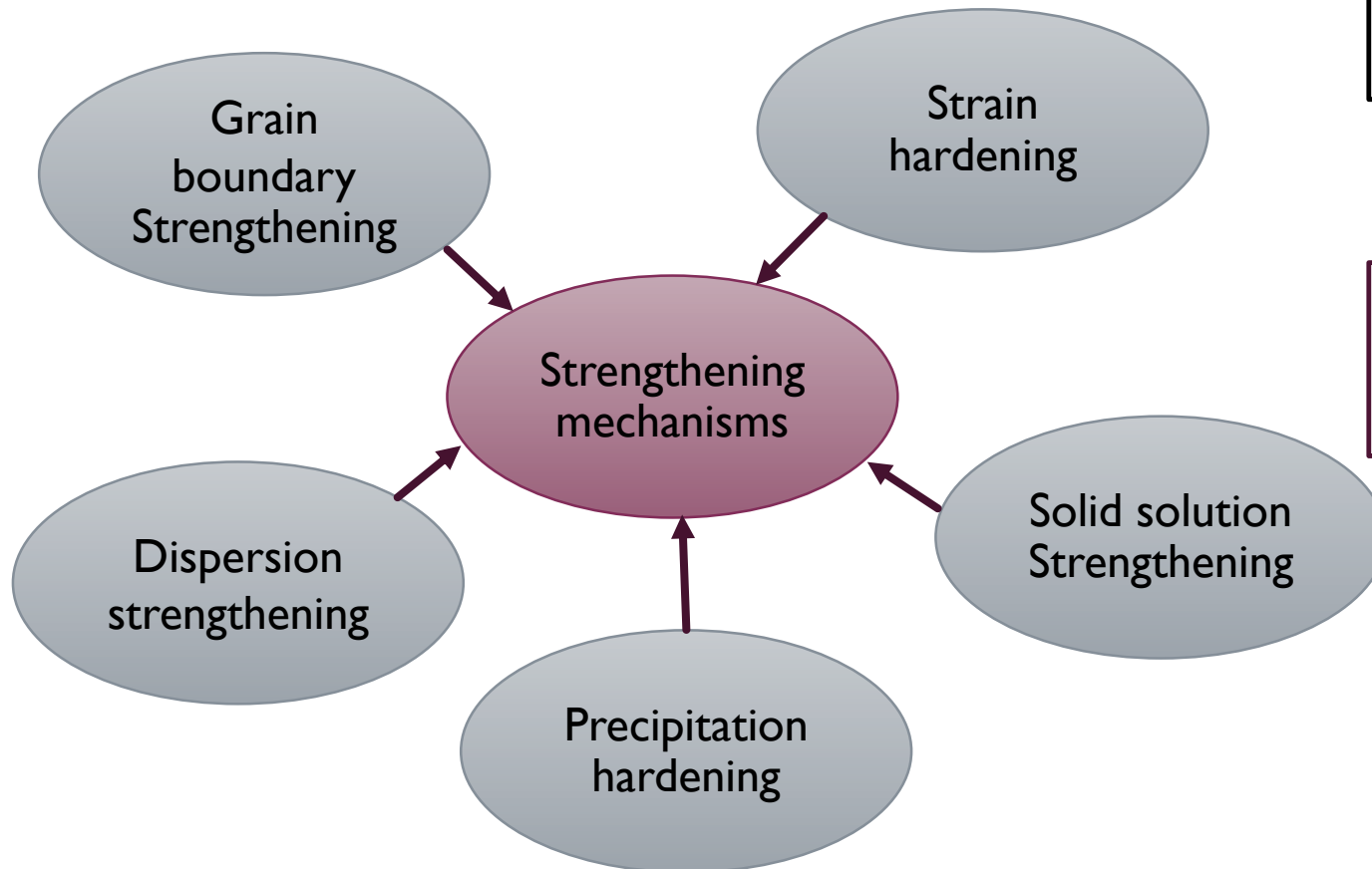
INTRODUCTION

Strengthening

- **The ability of a metal to deform plastically depends on the ability of dislocations to move.**
- **Hardness and strength are related to how easily a metal plastically deforms, so, by reducing dislocation movement, the mechanical strength can be improved.**
- **To the contrary, if dislocation movement is easy (unhindered), the metal will be soft, easy to deform.**

INTRODUCTION

Strengthening Mechanisms



The ability of a metal to plastically deform depends on the ability of dislocations to move.



Strengthening techniques rely on restricting dislocation motion to render a material harder and stronger.

To obtain material strength

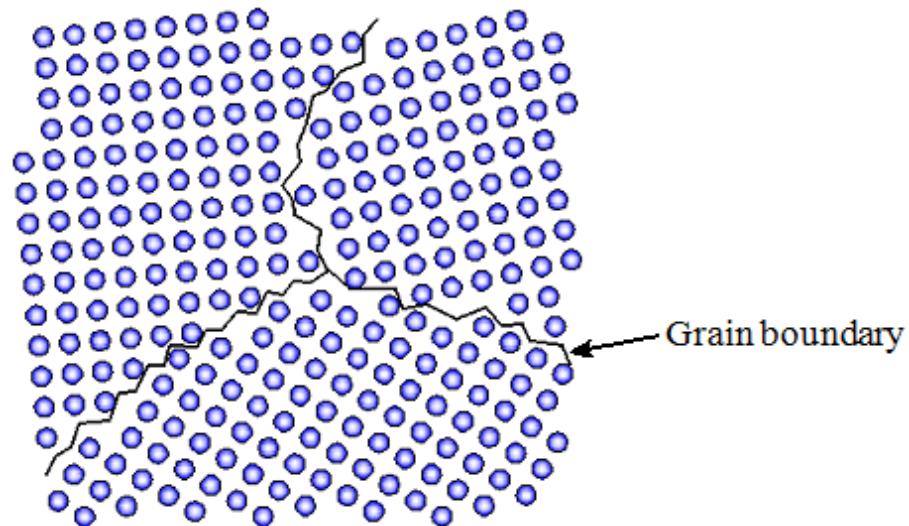


Sometimes ductility or toughness are decreased

GRAIN BOUNDARY STRENGTHENING

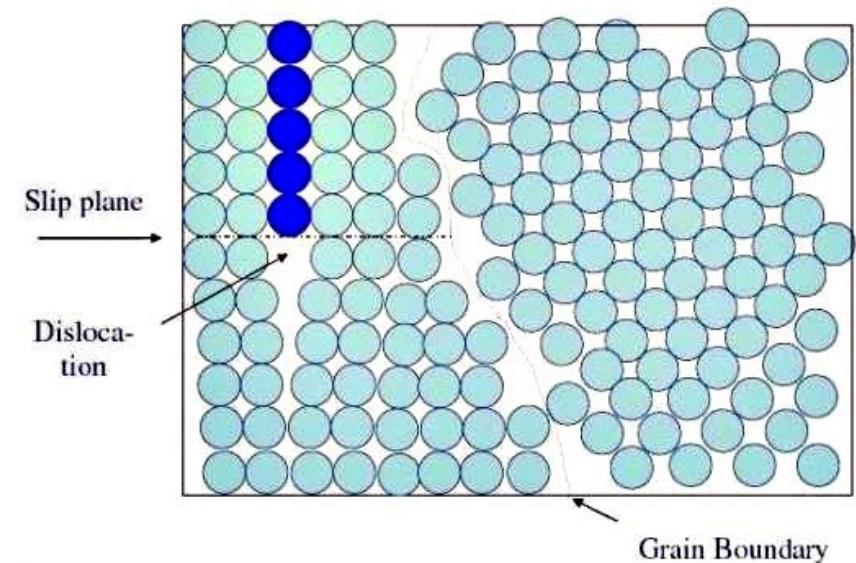
Grain boundaries

Grain boundary separates two grains having different crystallographic orientations.



Schematic atomic model of a grain boundary

Grain boundary structure contains grain boundary dislocations.



Motion of a dislocation as it encounters a grain boundary

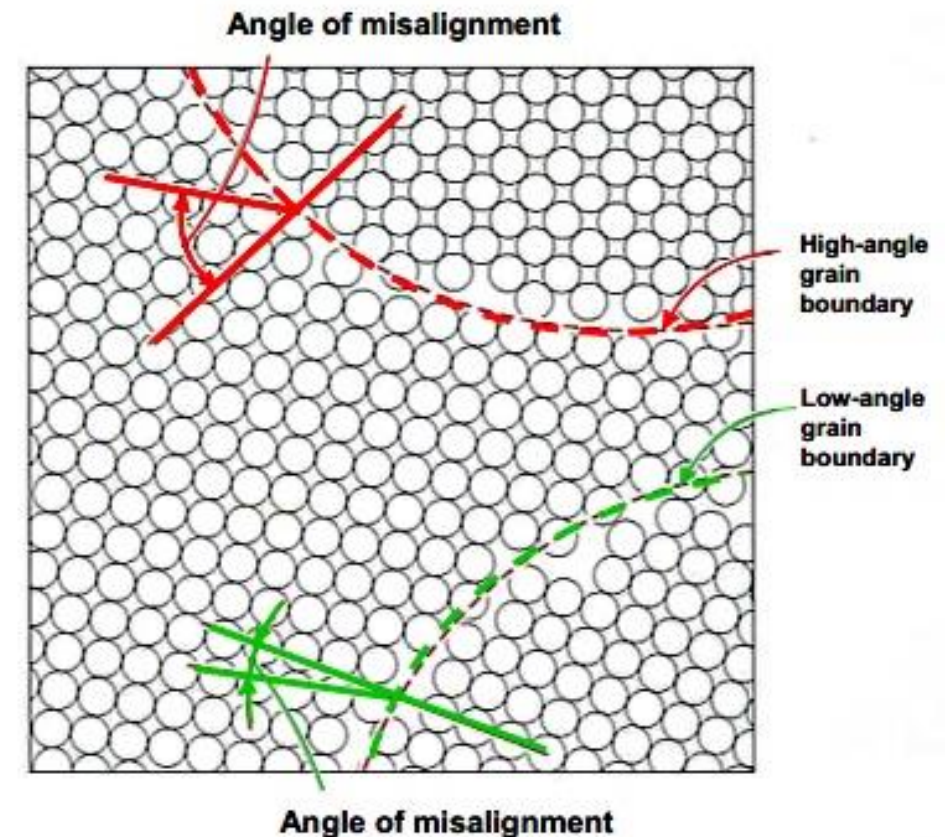
GRAIN BOUNDARY STRENGTHENING

High and low angle grain boundaries

High-angle grain boundary → high surface energy
Low-angle grain boundary → low surface energy

High energy grain boundary
serves as preferential sites
for solid state reactions;

- 1) Diffusion
- 2) Phase transformation
- 3) Precipitation

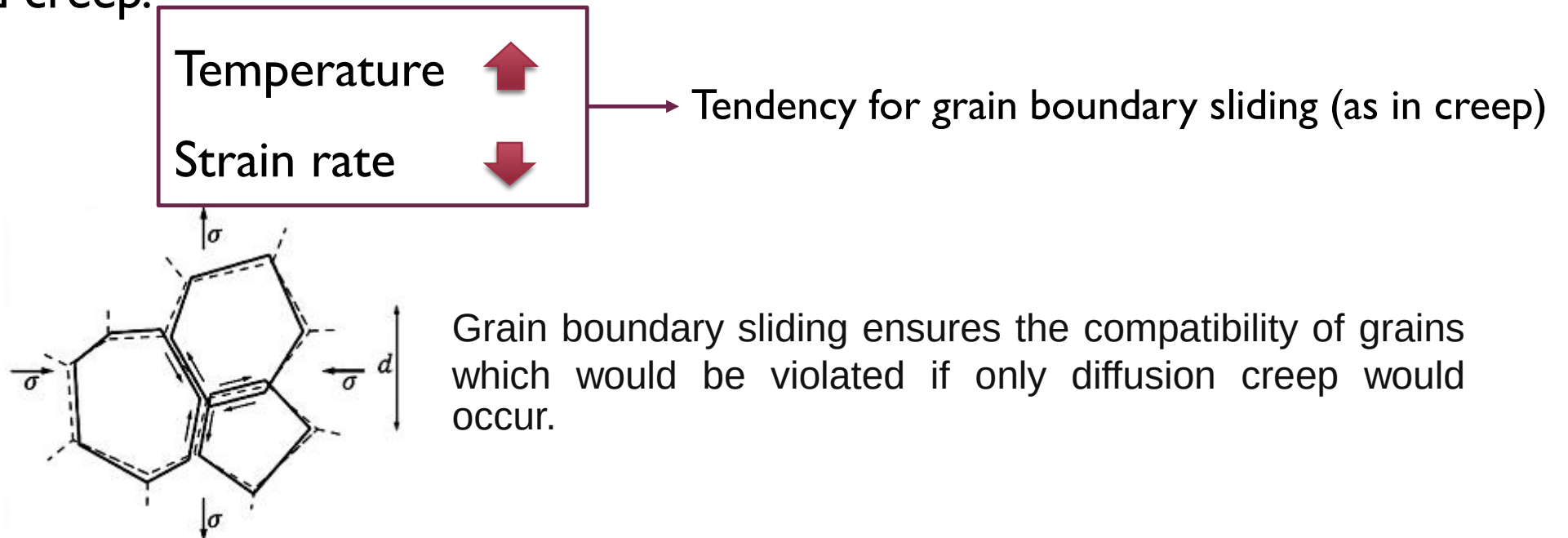


GRAIN BOUNDARY STRENGTHENING

Grain boundary sliding

At $T > 0.5T_m$, deformation can occur by sliding along the grain boundaries.

At high temperature; Grain boundary sliding is another possible mechanism for enhanced creep.

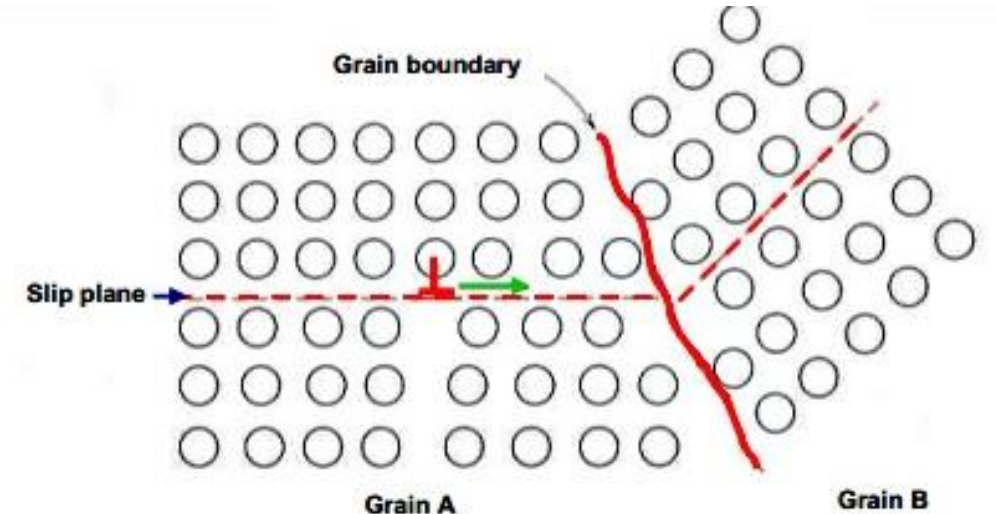


GRAIN BOUNDARY STRENGTHENING

Strengthening from grain boundaries

There are two important roles of the grain boundary which acts as a barrier to dislocation motion.

- 1) Difficulty for a dislocation to pass through **two different grain orientations** (need to change direction).
- 2) The atomic disorder within a grain boundary region contributes to a **discontinuity of slip planes from one grain to another**.



The motion of a dislocation as it encounters a grain boundary.

GRAIN BOUNDARY STRENGTHENING

Hall-Petch relation

A fine-grained material is harder and stronger than one that is coarse grained since greater amounts of grain boundaries in the fine-grained material impede dislocation motion.

The general relationship between the **yield stress** (tensile strength) and **grain size** was proposed by Hall and Petch.

$$\sigma_0 = \sigma_i + kD^{-1/2}$$

where σ_0 = the yield stress

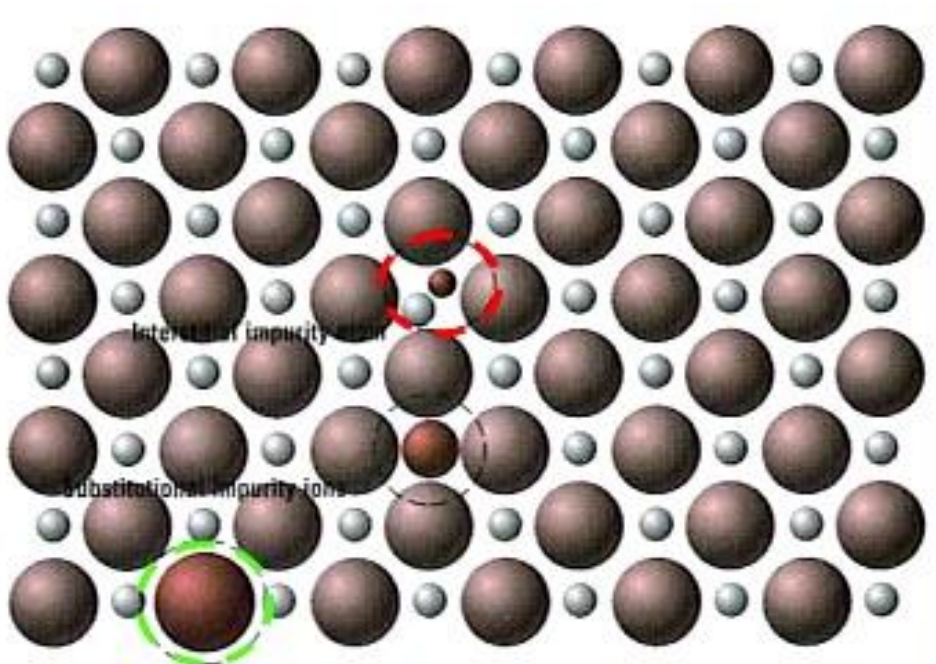
σ_i = the 'friction stress' or resistance to dislocation movement

k = constant value

D = grain diameter

SOLID SOLUTION STRENGTHENING

Solute atoms are introduced into the matrix (solvent atoms).



There are two types of solid solutions;

- 1) **Substitutional solid solution:** the solute and solvent atoms are **similar in size**, rendering the solute atoms to occupy lattice point of the solvent atoms.
- 2) **Interstitial solid solution:** The solute atoms are of **smaller size** than the solvent atom, rendering the solute atoms to occupy the interstitial sites in the solvent lattice.

SOLID SOLUTION STRENGTHENING

Factors affecting solubility of solute atoms

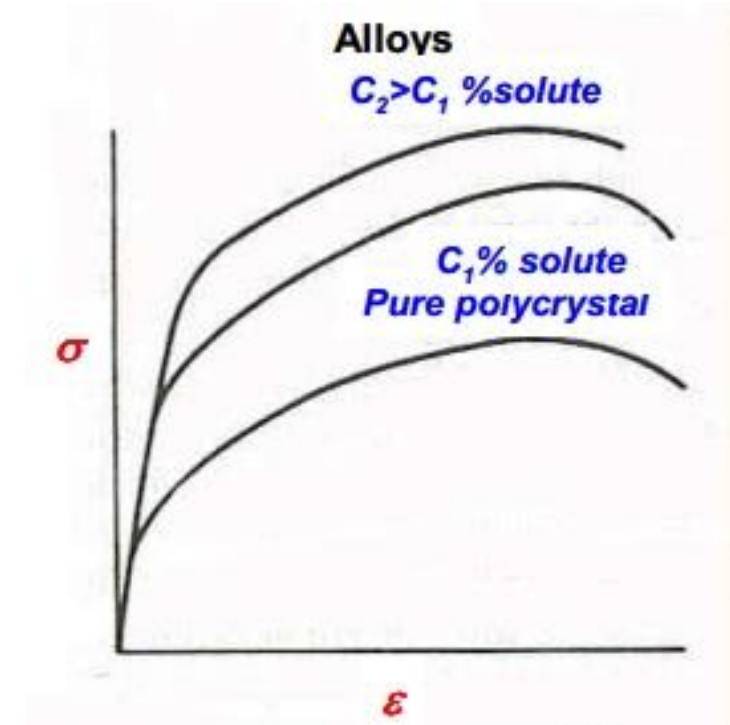
The **solubility of the solute atoms** in the host matrix (solvent) can be determined by several factors;

- 1) **Atomic size factors:** Solid solution is appreciable when the difference in atomic radii between the two atoms is $< \sim 15\%$, otherwise $\rightarrow$ creating substantial lattice distortion.
- 2) **Crystal structure:** **Similar crystal structure** of metals of both atom types are preferred.
- 3) **Electronegativity:** The more electropositive one element and the more electronegative the other, the more tendency to form an intermetallic compound than solid solution.
- 4) **Valences:** A metal will have more of a tendency to dissolve another metal of higher valency than one of a lower valency.

SOLID SOLUTION STRENGTHENING

Effects of solute alloy additions on stress-strain curve

- The addition of solute atoms raises the yield stress and the stress-strain curve as a whole.
- Therefore from Eq. $\sigma_0 = \sigma_i + kD^{-1/2}$
- The solute atoms should have more influence on the frictional resistance to dislocation motion σ_i than the locking of dislocation k .



Solute atoms



Strengthening effect

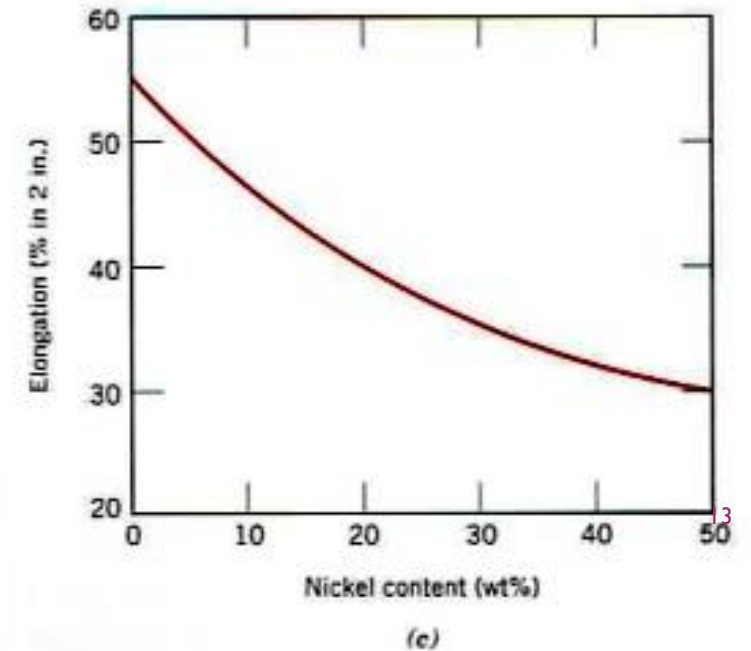
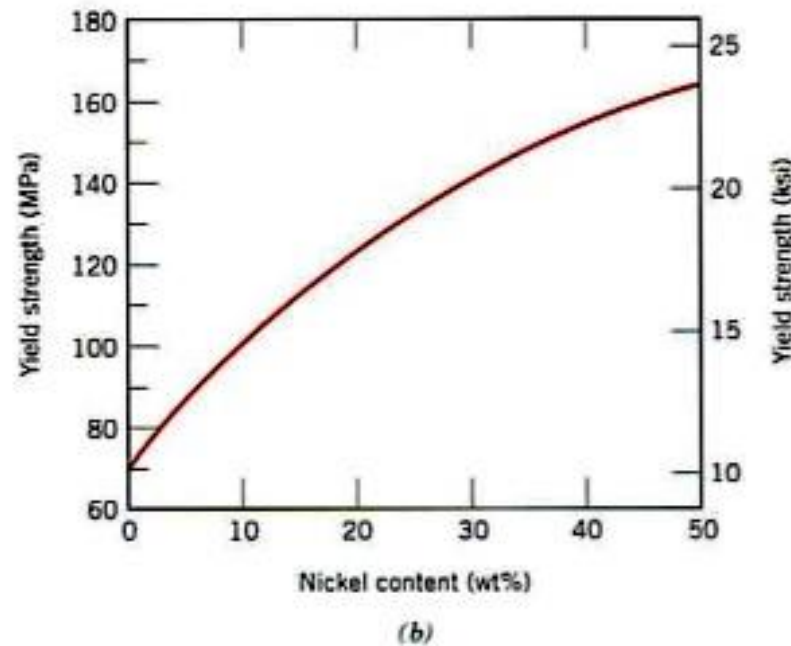
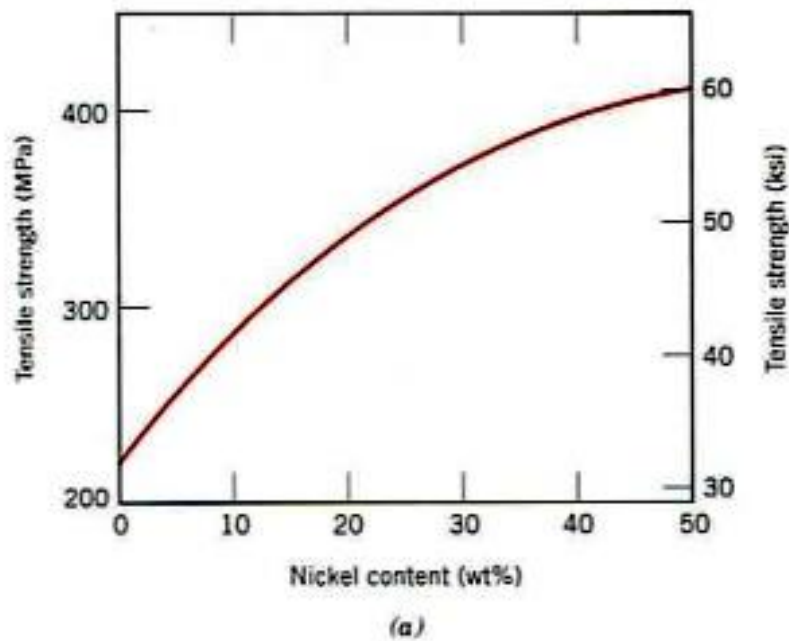


Effects of solute atoms on stress-strain curves.

SOLID SOLUTION STRENGTHENING

Effects of solute alloy additions on tensile properties

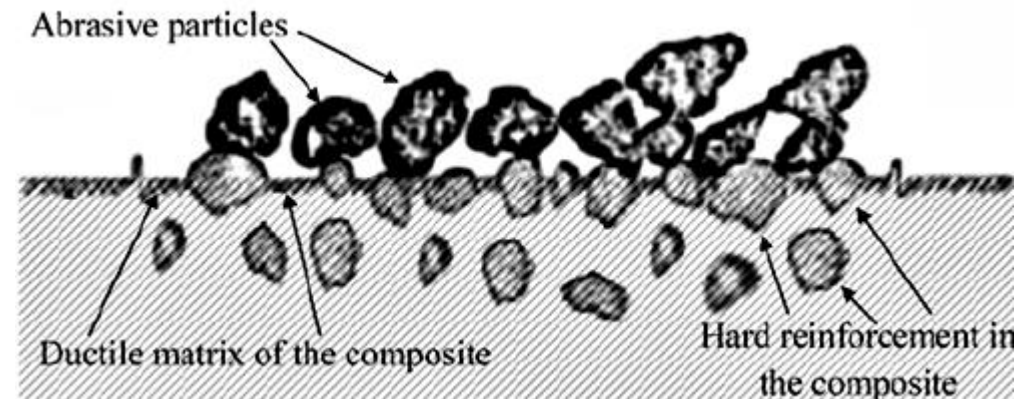
Impurity atoms ↑
Tensile strength ↑
Yield stress ↑
Ductility ↓



STRENGTHENING FROM SECOND PHASE

Many commercial alloys are composed of two or more metallurgical phases which provide strengthening effects:

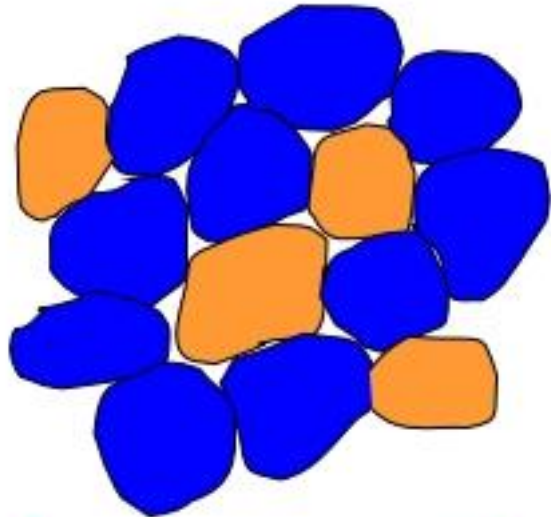
- Two phase aggregates
- Second phase/intermetallic particles
- Precipitation hardening



Strengthening from second phases is normally additive to the **solid solution strengthening** produced in the matrix.

STRENGTHENING FROM SECOND PHASE

Strengthening by two-phase aggregates



Two-phase aggregates

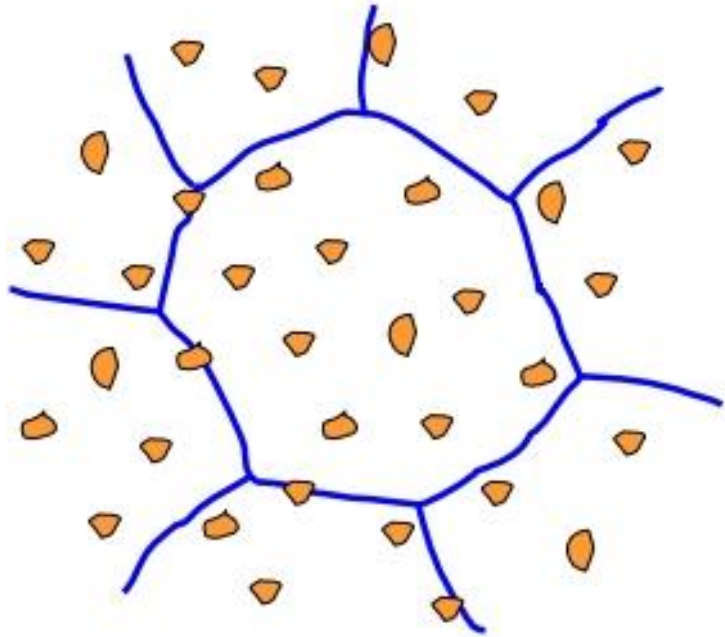
The size of the second phase particles are of similar size to that of the matrix.

Examples;

- Beta brass particles in an alpha brass matrix.
- Pearlite colonies in the ferrite matrix in annealed steels.

STRENGTHENING FROM SECOND PHASE

Strengthening by second phase particles



Dispersed second-phase particles in the matrix

- The **second phase or intermetallic particles** are much finer (down to submicroscopic dimensions) than the grain size of the matrix.
- The second phase particles produce **localized internal stresses** which alter the **plastic properties** of the matrix.

Factors influencing second-phase particle strengthening

Particle size

Particle shape

Distribution (interparticle spacing)

} **Strength**
Ductility
Strain hardening

STRENGTHENING FROM SECOND PHASE

Precipitation hardening

Precipitation hardening or age hardening requires the second phase which is **soluble at high temperature** but has a **limited solubility at lower temperatures**.

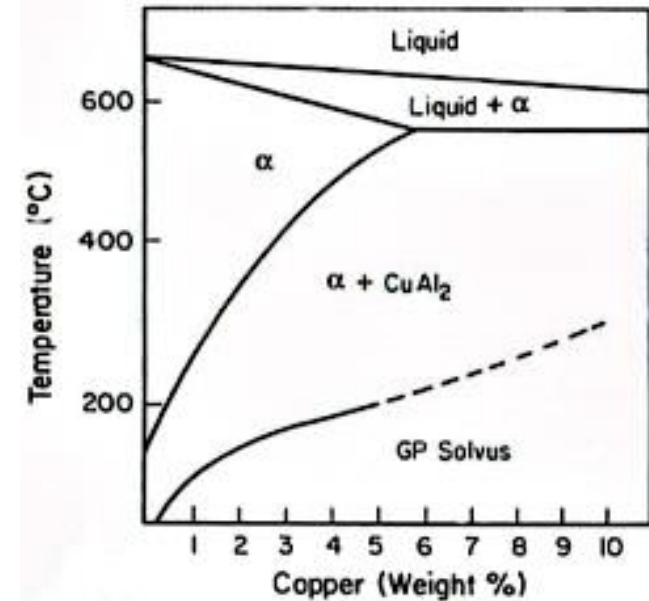
Solution treating at high temperature, then quenching

Second phase is in solid solution.

Ageing at low temperature

Precipitation of the second phase, giving strengthening effect.

Note: In precipitate-hardened system, there is coherency between the second-phase particle and the matrix.



Al-Cu phase diagram

Example: Age hardening aluminium alloys
Copper-beryllium alloys

STRENGTHENING FROM SECOND PHASE

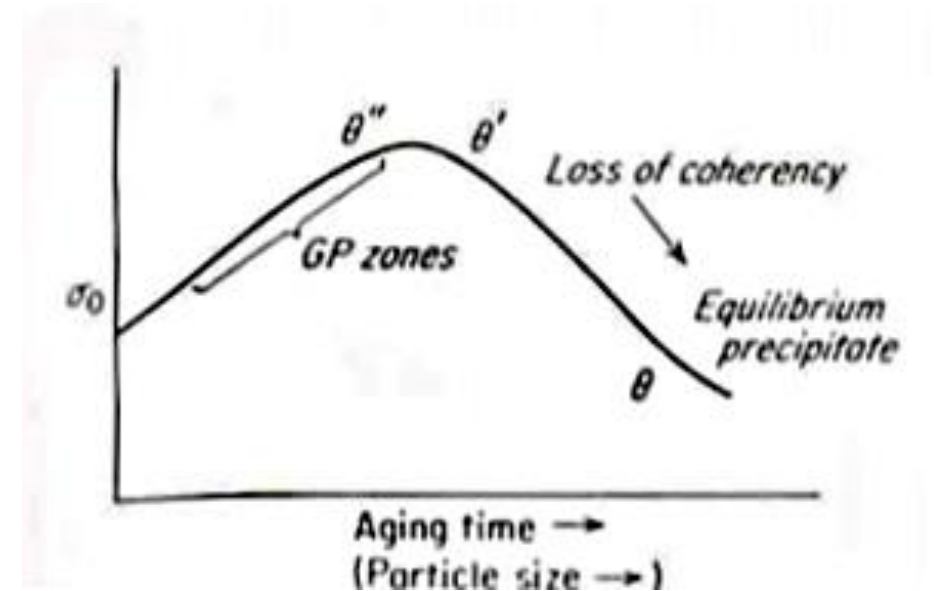
The formation of coherency precipitate

A number of steps occurs during precipitation hardening

- After quenching from solid solution the alloy contains **areas of solute segregation or clustering**. → **GP zone**.

This clustering is GP produces local strain giving higher hardness than the matrix.

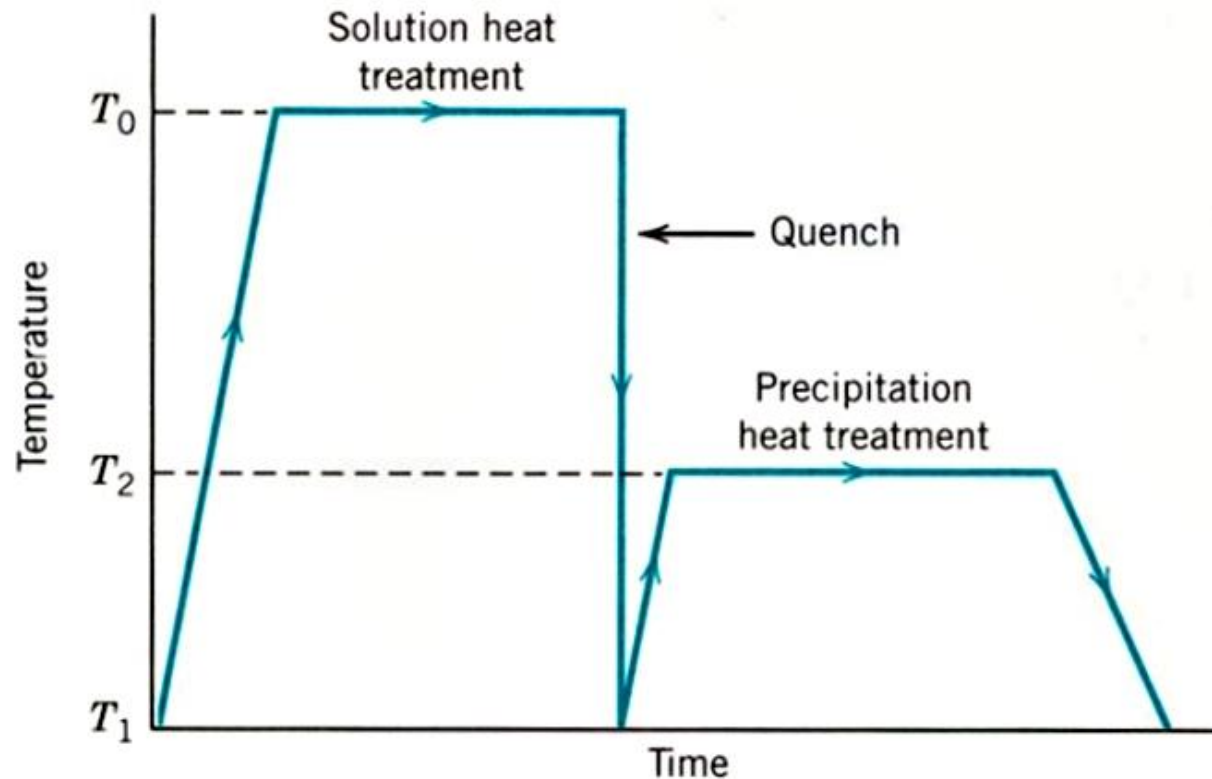
- The hardness of the **GP zone** increases with ageing time, developing GP or θ'' .
- Precipitate θ' is coherent with the matrix. → further **increase in hardness**.
- Further ageing produces θ , (**not coherent with the matrix**). → **lowering the hardness**.



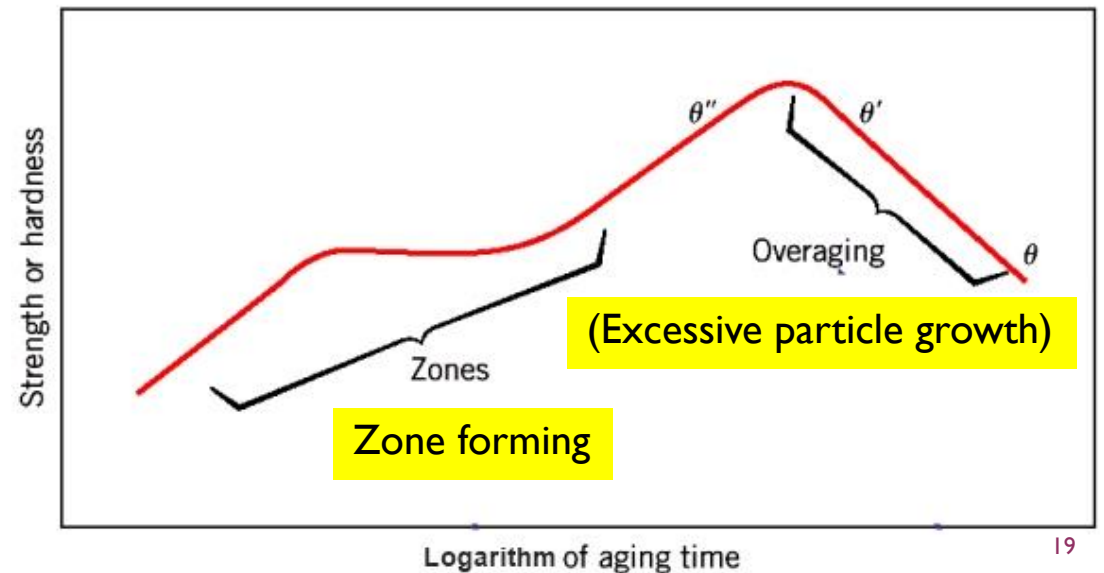
Variation of strength or hardness with ageing time

STRENGTHENING FROM SECOND PHASE

Heat Treatment Procedure

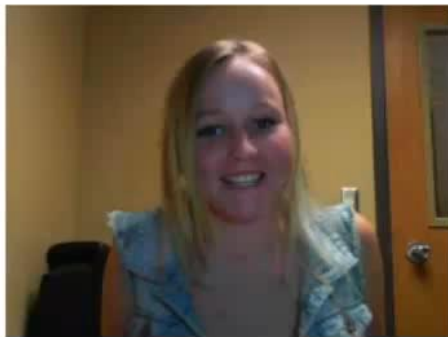


Property Change with aging Time



STRENGTHENING FROM SECOND PHASE

Precipitation Hardening



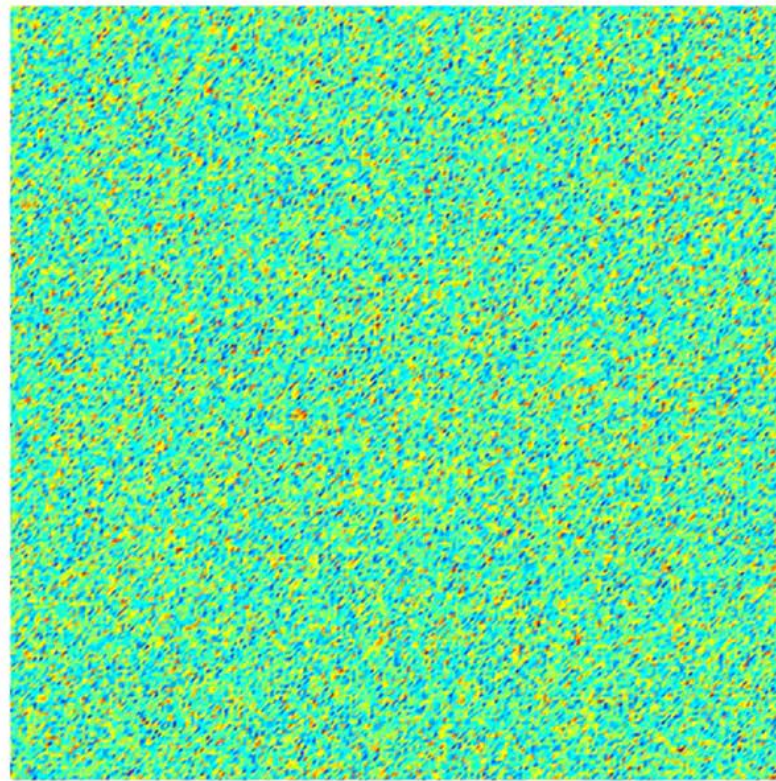
Oregon **TECH**

STRENGTHENING FROM SECOND PHASE

Three stages during aging

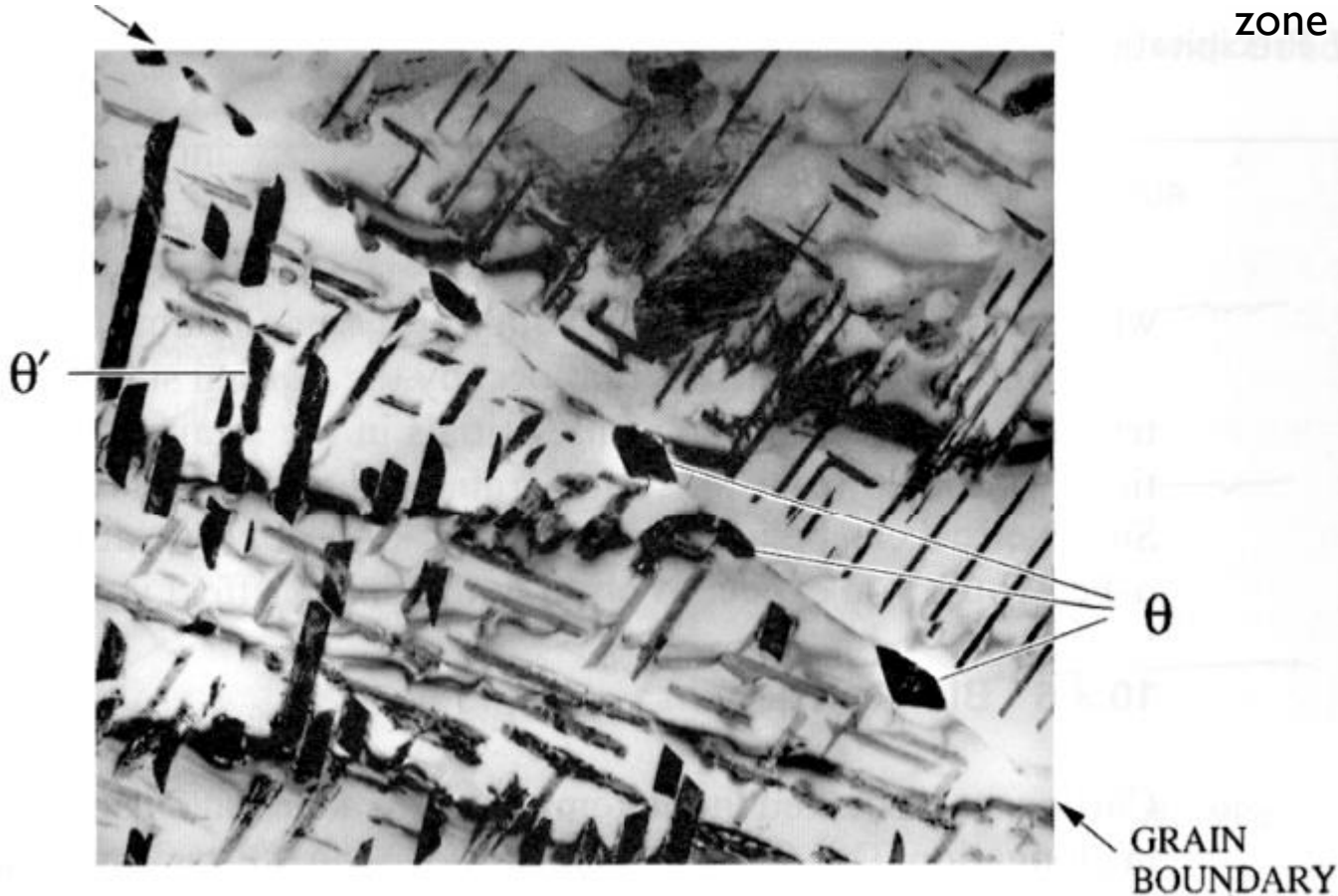
- **Stage I:** Clustering of solute atoms (Guinier Preston zones). Incubation period. The G-P zones are coherent with the matrix. (under-aged)
- **Stage II:** Nucleation and growth of second phase particles, until an equilibrium precipitate volume fraction is reached (peak-aged)
- **Stage III (Ostwald ripening):** Coarsening of precipitates, with large particles growing at the expense of small ones. (over-aged)

STRENGTHENING FROM SECOND PHASE

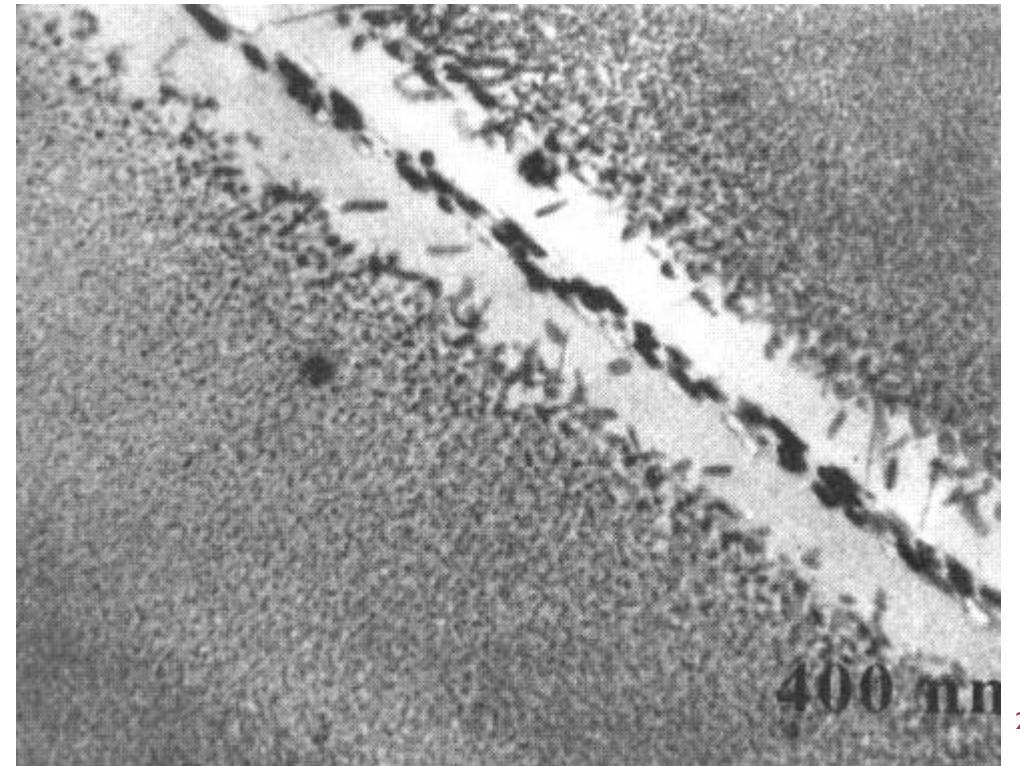


STRENGTHENING FROM SECOND PHASE

Precipitates in a Al-Cu alloy



Grain Boundary Precipitates and associated precipitate-free zone in an Al-Li alloy

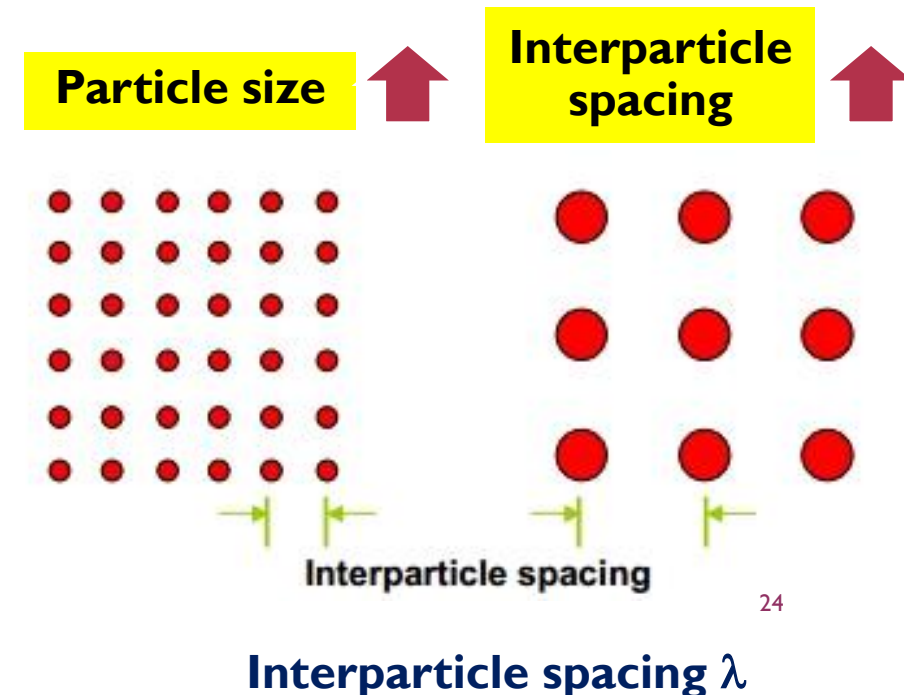


STRENGTHENING FROM SECOND PHASE

Factors affecting precipitation hardening

Particle size, shape, volume fraction and distribution are **key factors** in improving precipitation hardening.

- **High strength alloys** seem to consist of fine strong particles well distributed in deformed matrix.
- **Fine hard particles increase strength** by cutting dislocations → dislocation tangles → increasing strain hardening
- **Deformed matrix bares the load** which makes fracture more difficult.

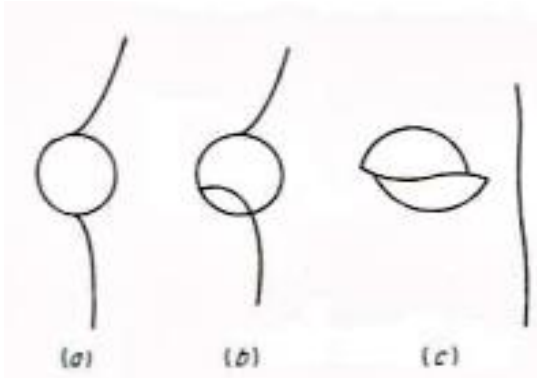


STRENGTHENING FROM SECOND PHASE

Interaction between fine particles and dislocations

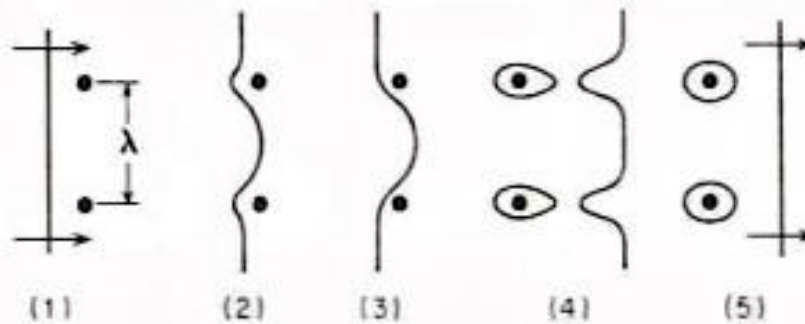
Second phase particles act in two distinct ways to retard the motion of dislocations.

1) Particles maybe cut by dislocation



- When the particles are small / soft

2) Particles allow dislocation to bypass/bow around them



- In over aged noncoherent precipitates. Bowing of dislocations around particles leaving dislocation loops behind

Orowan's mechanism of dispersion hardening

$$\tau_0 = \frac{Gb}{\lambda}$$

- Stress required to force dislocation between particles;

DISPERSION STRENGTHENING

Mechanism

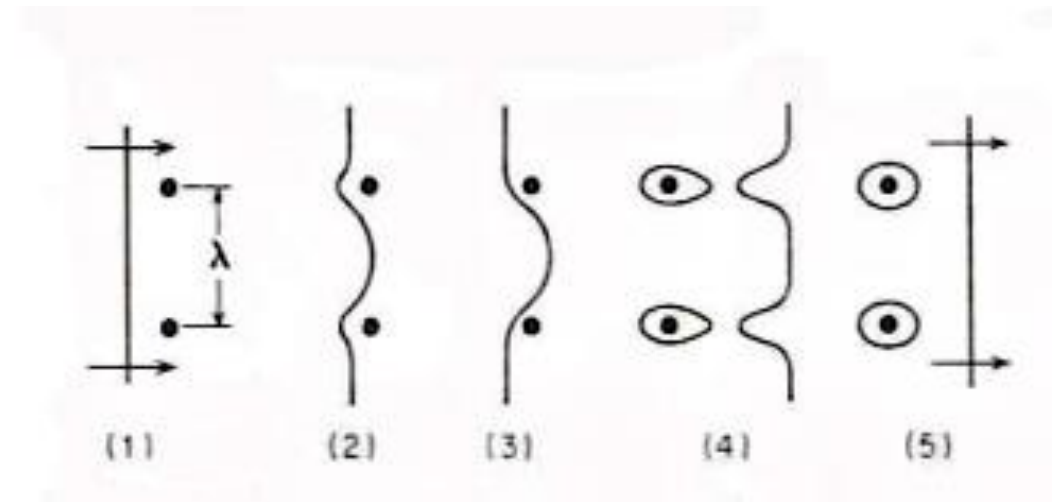
Orowan model

Dislocation bypass by the orowan bowing mechanism:

- The shear stress required to bow a two particles separated by λ

$$\tau_0 = \frac{Gb}{\lambda}$$

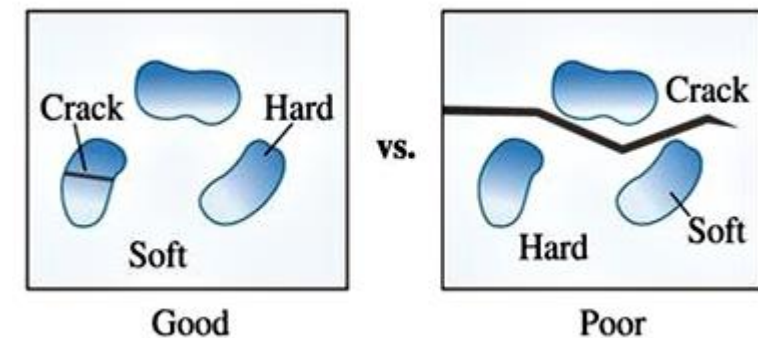
- Straight dislocation line approaching two particles.
- Initially line to bend.
- Reached to critical curvature.
- Leaving a dislocation loop around each particle.



DISPERSION STRENGTHENING

The strength and hardness in some metal alloys may be enhanced by the presence of extremely small and uniformly dispersed particles within the original phase matrix.

Whether introduced as insoluble particles in powder compaction (**dispersion strengthening**), or as precipitates in a solid state reaction (**precipitation or age hardening**), **second phase particles** are generally the most potent strengthening agent in practical high strength engineering materials.



Considerations for effective dispersion strengthening.
The precipitate phase should be hard and discontinuous.

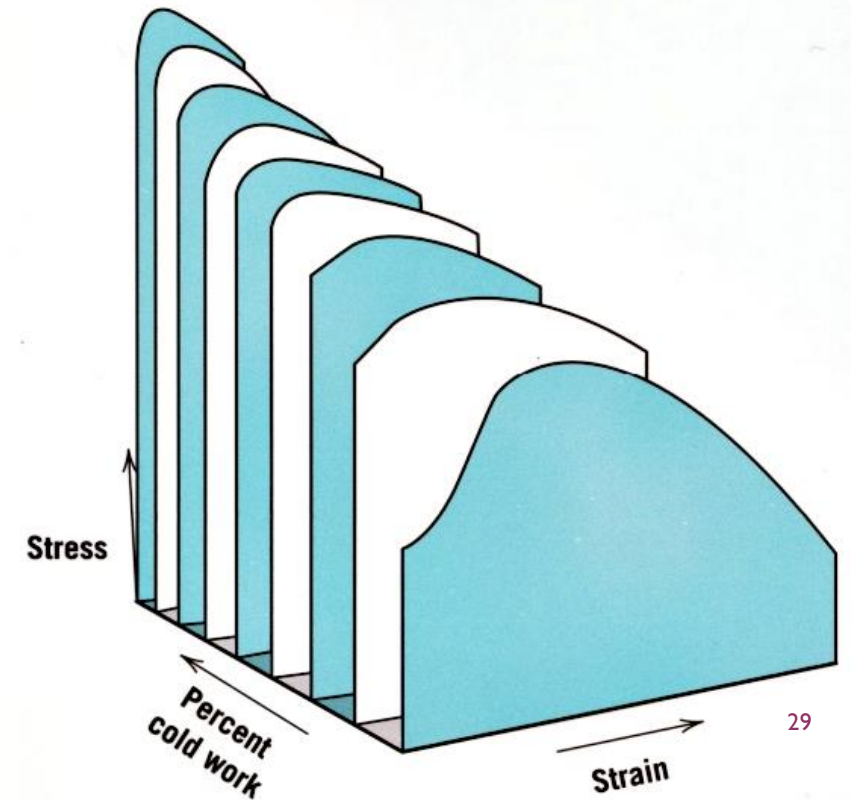
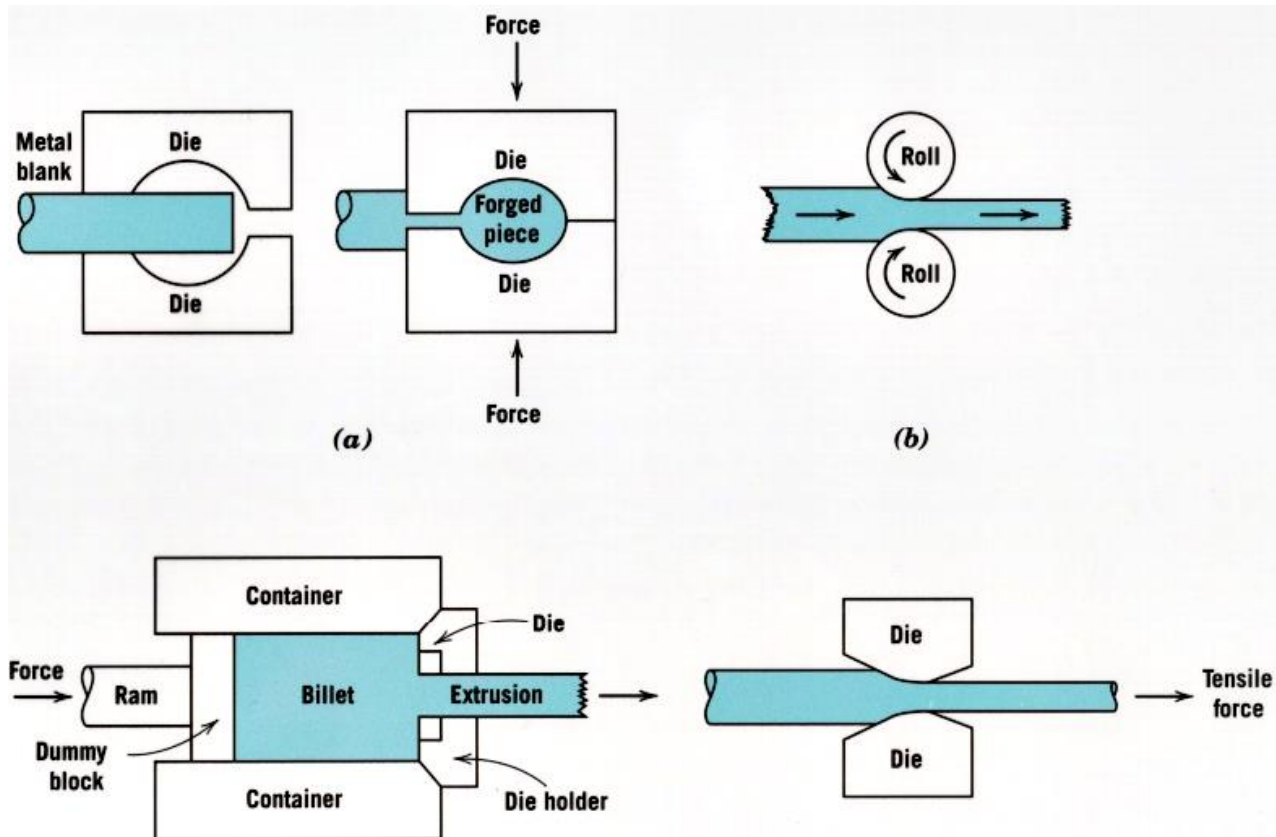
DISPERSION STRENGTHENING

Factor influencing second phase particle strengthening

- Interparticle spacing (λ)
- Average particle diameter (D)
- Volume fraction (V_f)
- Distribution

STRAIN HARDENING OR COLD WORKING

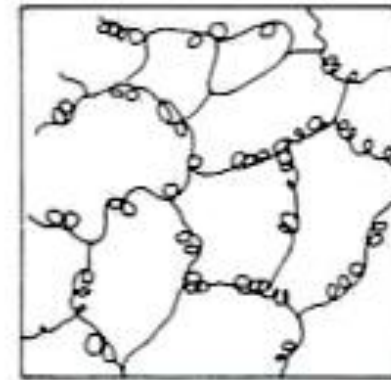
- Room temperature deformation
- Common forming techniques used to change the cross sectional area:



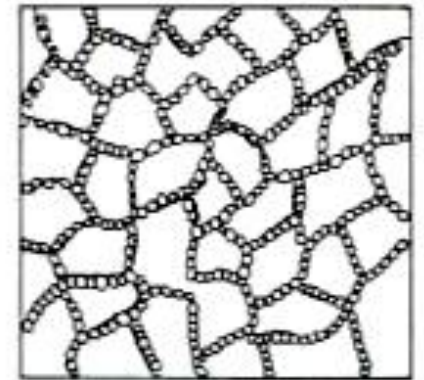
STRAIN HARDENING OR COLD WORKING

Cold-work structure occurs when plastic deformation carried out at in a temperature region and over a time interval such that the strain hardening is not relieved.

- **Cold worked structure** contains dislocation $\sim 10^{11}$ mm^{-2} , while annealed structure possesses $\sim 10^4$ to 10^6 mm^{-2} .
- As the deformation proceeds, the high density dislocations tangles form the **cell walls**.
- About 10% of energy input in cold work process is stored in the lattice



(a) Deformed to 10%



(b) Deformed to 50%

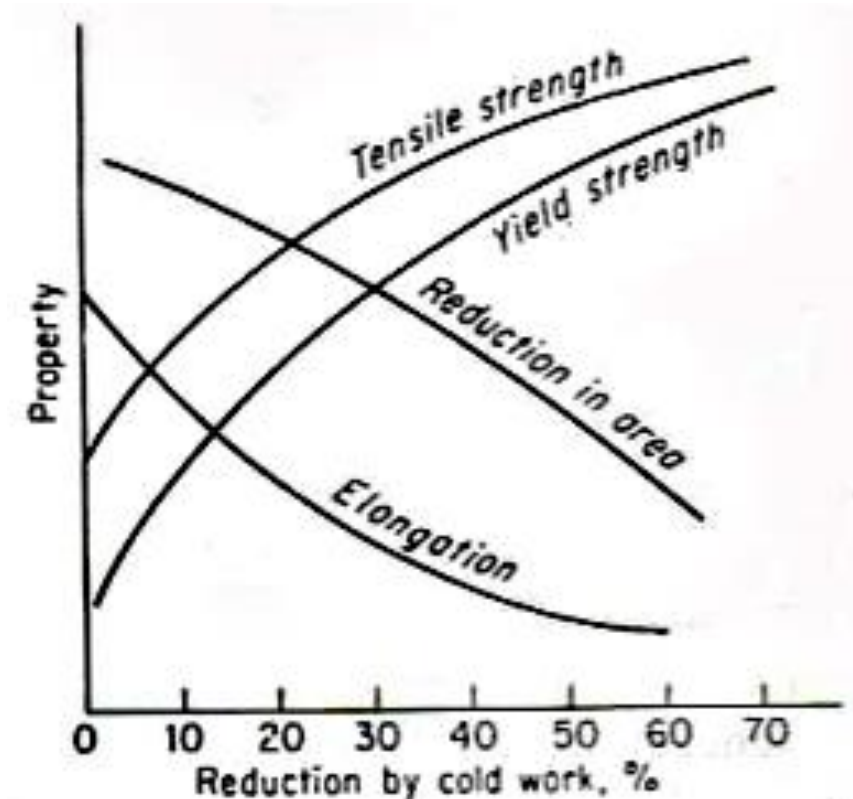
Dislocations in cell walls

STRAIN HARDENING OR COLD WORKING

Strain hardening or cold working is used to harden alloys that do not respond to heat treatment.



- The rate of strain hardening is lower in **HCP** than in **cubic metals**.
- The final strength of **cold-worked solid solution alloy** is almost always greater than that of the **pure metal cold-worked to the same extent**.



STRAIN HARDENING OR COLD WORKING

Dislocations during cold working



- Dislocations entangle one another during cold work.
- Dislocation motion becomes more difficult, which makes the material stronger overall.

Result of Cold Working

- Dislocation density increases, which leads to an increase in yield strength: Materials become harder.
- Ductility decreases and tensile strength also increases.

